



EMMTE: An Excel VBA tool for source apportionment of nitrate based on the stable isotope mixing model



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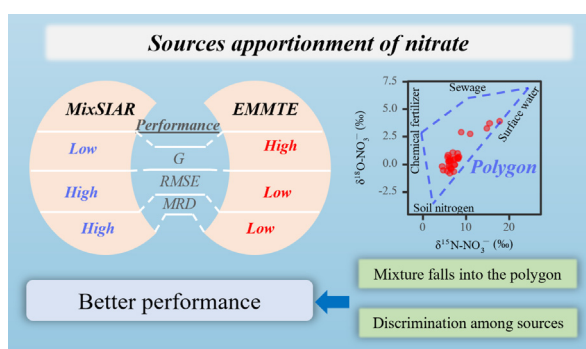
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HIGHLIGHTS

- EMMTE, one Excel VBA tool for source apportionment of nitrate was developed.
- There was high consistency in source apportionments between EMMTE and MixSIAR.
- The performances of EMMTE were even better than MixSIAR in most cases.
- Mixture's isotopic signature within the polygon delineated by sources determined the performance of EMMTE.

GRAPHICAL ABSTRACT



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ABSTRACT

Dual nitrate stable isotopes combined with end-member mixing models are typically used to identify nitrate sources in fields of geochemistry and environmental science, which helps to quantitatively depict the geochemical behaviors of nitrate and accurately control the sources of nitrate pollution in waters. Recently, various models with different computation principles, working efficiency, and operation difficulty have been developed and applied in the source apportionment of nitrate. In this paper, an end-member mixing model tool on Excel™, namely EMMTE, has been written with Visual Basic for Application (VBA) and built into a macro-enabled Excel™ spreadsheet. Monte Carlo simulation and constraint relative deviation between the observed and the predicted values were included in the working algorithm to solve the mass balance equation. After comparison with the internationally recognized Bayesian framework (mixing stable isotope analysis in R, MixSIAR) in different cases (three practical cases and one virtual case), the preliminary results showed that the contribution of various sources to groundwater nitrate calculated by EMMTE was highly consistent with that by MixSIAR and the performance of EMMTE seemed to be as good as that of MixSIAR as indicated by the higher goodness-of-prediction, lower root-mean-square error, and lower relative deviation. Therefore, EMMTE is applicable in the source apportionment of groundwater nitrate, and might also be extended to other water bodies and mixtures. It provides a simple, feasible, and user-friendly for front-line workers without experience with MixSIAR to quantitatively source apportionment of nitrate in waters.

1. Introduction

In recent decades, nitrate has been continuously threatening the supply of drinking water sources, and further brings a considerable risk to human

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health throughout the world (Cai et al., 2022; Cao et al., 2022; Carrey et al., 2021; Rivett et al., 2008). Therefore, it is critical to qualitatively identify the major sources for the well-being of human society and provide significant insights into improving the water quality (Torres-Martinez et al., 2020; Xiong et al., 2021; Zhang et al., 2020). Moreover, quantifying the contribution of various nitrogen sources and accurately control nitrate pollution in various water bodies, especially in the groundwater, is significant in advancing our knowledge of the nitrogen cycle across complicated hydrogeology conditions and a series of biogeochemical processes (Niu et al., 2022; Xiong et al., 2021; Zhao et al., 2019).

Due to the distinct isotopic compositions of sources and their predictable and recognizable changes caused by biogeochemistry cycles, isotopes can offer a direct way to identify sources of nitrate in circumstances (Kendall, 1998; Nikolenko et al., 2018; Xue et al., 2009). Kohl et al. (1971) first used nitrate nitrogen isotope ($\delta^{15}\text{N-NO}_3^-$) to estimate the contribution of fertilizer NO_3^- -N in the Sangamon River (Illinois, USA), subsequently the stable isotopes of $\delta^{15}\text{N-NO}_3^-$ and oxygen ($\delta^{18}\text{O-NO}_3^-$) are recognized as powerful tools to track nitrate sources and have been successfully and widely applied in groundwater (Abass et al., 2020; Liu et al., 2006; Minet et al., 2017; Panno et al., 2001), rivers (Bu et al., 2017; Li et al., 2014; Liu et al., 2013), reservoirs (Jin et al., 2019; Zhang et al., 2018; Zhao et al., 2019), lakes (Ding et al., 2014; Hu et al., 2019; Romanelli et al., 2020), and stormwater runoff (Yang and Toor, 2016).

To quantitatively apportion sources of nitrate by dual nitrate isotopes, end-member mixing models with various carriers, assumptions, algorithms, applicability, and working efficiency were developed and summarized in Table 1 (Stock et al., 2018; Wang et al., 2019). As the principle of the end-member mixing model (isotope mixing model), the simple mass balance equation of dual nitrate stable isotopes was the fundamental solution when no more than three nitrate sources were included in a determinate system of three equations and three unknowns (Li et al., 2014; Liu et al., 2014). However, it is limited to no more than three sources, neglects source variability and isotopic fractionation, and has low work efficiency without batch calculation (Table 1) (Cao et al., 2022; Xue et al., 2012; Xue et al., 2009). Subsequently, *IsoSource*, with a deterministic algorithm for providing approximate solutions to represent the solution polytope, was developed and proposed by US Environmental Protection Agency (EPA) (Phillips and Gregg, 2003). Although this tool has a graphical user interface (GUI) and can calculate the contribution of maximum ten sources to the mixture, however, the isotopic signature of the mixture should be within the polygon determined by sources (Begum et al., 2022; Phillips and Gregg, 2003). Besides, it does not consider source variability, isotopic fractionation, and batch calculation (Begum et al., 2022; Ji et al., 2017; Jin et al., 2019). To solve the indeterminate system and consider the uncertainty, the stable isotope analysis in R (SIAR) employed an algorithm to determine the probability distributions for proportional contributions based on a Bayesian framework (Moore and Semmens, 2008; Parnell et al., 2010). However, running this tool needs to master R language, which is inconvenient for beginners (Cao et al., 2022). Subsequently, the mixing SIAR (*MixSIAR*) with GUI was collaboratively developed based on *MixSIR* and SIAR (Liu and Han, 2021; Stock et al., 2018; Stock and Semmens, 2016a). Although *MixSIAR* has many advantages and is considered the most

popular and acceptable model (Table 1), its GUI in R requires loading the dependency packages of gWidgetsRGtk2, RGtk2, gWidgets, and cairoDevice (Stock and Semmens, 2016a), which are complicated to install and unfriendly to researchers who are not familiar with R.

Referring to the advantages and disadvantages of the existing models, we developed an end-member mixing model tool on Excel™, namely *EMMTE*, based on the stable isotope mixing model. *EMMTE* has been written with Visual Basic for Application (VBA) and built into a macro-enabled Excel™ spreadsheet. In the study, we selected three practical cases from previous studies and one virtual case that have different sources and used *MixSIAR* and *EMMTE* to calculate the contribution of various sources to groundwater nitrate, respectively. The study aims to (1) evaluate the applicability and reliability of *EMMTE* in calculating the contribution of potential sources to nitrate and (2) compare the performance of *MixSIAR* and *EMMTE* for source apportionment of groundwater nitrate. It is wished to provide some convenience for front-line researchers who apportion the contribution of nitrate from various sources.

1.1. Representation of dual nitrate isotopes

The stable isotope abundances (δ_{sample}) of dual nitrate isotopes are expressed in parts per thousand as follows:

$$\delta_{\text{sample}}(\text{‰}) = \left(\frac{R_{\text{sample}}}{R_{\text{standard}}} - 1 \right) * 1000 \quad (1)$$

where R_{sample} and R_{standard} are the $^{15}\text{N}/^{14}\text{N}$ or $^{18}\text{O}/^{16}\text{O}$ ratios of the samples and standards for $\delta^{15}\text{N-NO}_3^-$ and $\delta^{18}\text{O-NO}_3^-$, respectively. The $\delta^{15}\text{N-NO}_3^-$ values are reported relative to atmospheric N_2 and the $\delta^{18}\text{O-NO}_3^-$ values are reported relative to Vienna Standard Mean Ocean Water (VSMOW).

1.2. Description of MixSIAR and EMMTE

MixSIAR is one Bayesian framework with Markov chain Monte Carlo (MCMC) built on R, more information refers to Stock and Semmens (2016a, 2016b), and Stock et al. (2018). Based on the mass balance model and Monte Carlo random simulation, we developed *EMMTE* in Excel™ using VBA, the basic principles can be expressed as:

$$\delta^{15}\text{N}_{m,k} = \sum_{n=1}^N f_{s,n,k} \delta^{15}\text{N}_{s,n} + \delta^{15}\text{N}_{\text{error},k} \quad (2)$$

$$\delta^{18}\text{O}_{m,k} = \sum_{n=1}^N f_{s,n,k} \delta^{18}\text{O}_{s,n} + \delta^{18}\text{O}_{\text{error},k} \quad (3)$$

$$1 = \sum_{n=1}^N f_{s,n,k} \quad (4)$$

$$RD_{\delta^{15}\text{N}} = \left| \frac{\delta^{15}\text{N}_{\text{error},k}}{\delta^{15}\text{N}_{m,k}} \right| \times 100 \leq RD \quad (5)$$

Table 1
Inter-comparison of commonly stable isotope mixing models.

Tools	GUI	Carrier	Maximum sources	Source variability	Isotopic fractionation	Batch calculation	Writing code	Visualization on map	Convenient operation	References
Mass balance model	×	Conceptual model	3	×	×	×	×	×	√	(Cao et al., 2022; Xue et al., 2012; Xue et al., 2009)
<i>IsoSource</i>	√	Independent software	10	×	×	×	×	×	√	(Begum et al., 2022; Phillips and Gregg, 2003; Wang et al., 2019)
SIAR	×	SIAR package in R	>3	√	√	√	√	×	×	(Ji et al., 2017; Moore and Semmens, 2008; Xue et al., 2009)
<i>MixSIAR</i>	√	<i>MixSIAR</i> package in R	>3	√	√	√	√	×	×	(Cao et al., 2022; Liu and Han, 2021; Stock et al., 2018)
<i>EMMTE</i>	√	Excel VBA	10	×	×	√	×	√	√	This study

$$RD_{\delta^{18}O} = \left| \frac{\delta^{18}O_{error,k}}{\delta^{18}O_{m,k}} \right| \times 100 \leq RD \quad (6)$$

where $\delta^{15}N_{m,k}$ and $\delta^{18}O_{m,k}$ are $\delta^{15}N-NO_3^-$ and $\delta^{18}O-NO_3^-$ in the mixture (groundwater) for the k^{th} sampling, respectively; $\delta^{15}N_{error,k}$ and $\delta^{18}O_{error,k}$ are the residual error between the observed abundances and the predicted abundances at the k^{th} samplings of the total of K samplings for $\delta^{15}N-NO_3^-$ and $\delta^{18}O-NO_3^-$, respectively; $\delta^{15}N_{s,n}$ and $\delta^{18}O_{s,n}$ are $\delta^{15}N-NO_3^-$ and $\delta^{18}O-NO_3^-$ for the n^{th} sources, respectively; $f_{s,n,k}$ is the proportion of the n^{th} source for the k^{th} sampling in the mixture (groundwater); relative deviation (RD) is the given to screening the random samplings. In our algorithm, we also offer parameters of increment and the minimum number of feasible solutions ($mins$) to stop the calculation. In brief, during a running of K -samplings for a specific mixture, the program will not stop until the number of solutions is larger than the given $mins$ among which both $RD_{\delta^{15}N}$ and $RD_{\delta^{18}O}$ are lower than RD . If there were no acceptable solutions during the 1st running of the K -samplings, the second or more running of K -samplings would be conducted with the added RD by the increment of RD .

EMMTE is comprised of four sheets, namely Parameter Setting, Data Enter, Results, and Map, and one chart with the name of Mixture and Sources (Fig. 1). More detailed description refers to Text S1 of supplementary materials (SM) in Appendix A. The simple operation procedure of *EMMTE* refers to Text S2 of SM, and *EMMTE* can be obtained in the Appendix A.

1.3. Applicability test by various cases

A series of potential sources, including atmospheric precipitation (AP), chemical fertilizer (CF), manure and sewage (MS), soil nitrogen (SN), and

surface water (SW), may contribute nitrate to groundwater (Cao et al., 2022; Kendall et al., 2007; Xue et al., 2009; Yang et al., 2020). There were three practical cases and one virtual case with different source compositions were selected based on previous studies. The first three cases included three, four, and five sources of nitrate, and the data was obtained from Ming et al. (2020), Cao et al. (2022), and Zhang et al. (2020), respectively. The fourth case had six sources of nitrate and the data was randomly generated by the random function of RANDBETWEEN() in Excel™.

1.4. Operational settings for MixSIAR and EMMTE

In *MixSIAR* (*MixSIARgui* 1.0.0), the dual nitrate isotopes of the mixture (groundwater), the source data (Table S1), and discrimination (set to zero) were loaded. The fractionation of the dual nitrate isotopes was not considered because there was no significant denitrification in the selected cases (Cao et al., 2022; Ming et al., 2020; Zhang et al., 2020). To have consistency with *EMMTE*, we loaded the mean value of sources as the “raw data” at the procedure of “Load source data”. The MCMC run length, error structure, and specify prior were selected as normal, Resid * Process, and Uninformative/Generalist, respectively (Stock and Semmens, 2016a; Stock and Semmens, 2016b). In *EMMTE*, the mean isotope abundances of sources were entered, and simulation times were set to 1×10^4 , 1×10^5 , 1×10^6 , and 1×10^7 in the four cases, respectively. The increment, RD , and minimum number of feasible solutions were set to 1, 0, and 10, respectively.

1.5. Performance assessment

Before the performance assessment of *MixSIAR* and *EMMTE*, the predicted abundances of dual nitrate isotopes for each sample ($P_{i,g}$) were



Fig. 1. The graphical user interface (GUI) and operation procedure of *EMMTE*.

calculated following the assumption that the composition of dual nitrate isotopes of each sample was a mixture of various potential sources (Eq. (7)). In detail, after obtaining the contribution fraction of the h^{th} source for the i^{th} sample in terms of the g^{th} isotope ($CF_{h,i,g}$) by *MixSIAR* or *EMMTE*, we could calculate the abundance of dual nitrate isotopes contributed by the h^{th} source for the i^{th} samples through the product of the $CF_{h,i,g}$ and the g^{th} nitrate isotope (average) in the h^{th} sources ($S_{h,g}$). Then, we calculated the sum of the abundance of dual nitrate isotopes contributed by all sources. According to the mass balance principle, we assumed that the sum was the predicted abundance of dual nitrate isotopes for each sample.

$$P_{i,g} = \sum_{h=1}^H CF_{h,i,g} * S_{h,g} \quad (g = 1, 2) \quad (7)$$

where H is the total number of sources; g denotes the individual nitrate isotope $\delta^{15}\text{N}-\text{NO}_3^-$ or $\delta^{18}\text{O}-\text{NO}_3^-$ when it is equal to 1 or 2.

The performance of *MixSIAR* and *EMMTE* was evaluated through the matching degree between the observed ($O_{i,g}$) and $P_{i,g}$ of dual nitrate isotopes through Eqs. (8)–(10). The goodness-of-prediction (G_g) was used to determine the applicability of the two models in source apportionment of groundwater nitrate. The closer the G_g value to 1, the higher the reliability of the model; a negative G_g value indicates that the model is less reliable (Wang et al., 2019).

$$G_g = 1 - \left[\frac{\sum_{i=1}^N (P_{i,g} - O_{i,g})^2}{\sum_{i=1}^N (O_{i,g} - \bar{O}_g)^2} \right] \quad (g = 1, 2) \quad (8)$$

where $\bar{O}_g$ is the average of $O_{i,g}$.

The root-mean-square error (RMSE) was used to evaluate the overall prediction effect of dual nitrate isotopes as it summarizes the average difference between the $O_{i,g}$ and $P_{i,g}$ (Li and Heap, 2011).

$$RMSE_g = \left[\frac{1}{N} \sum_{i=1}^N (P_{i,g} - O_{i,g})^2 \right]^{\frac{1}{2}} \quad (g = 1, 2) \quad (9)$$

The median of RD (MRD_g) between $O_{i,g}$ and $P_{i,g}$ was used to test the accuracy of each isotope as follows:

$$MRD_g(\%) = \text{Mdn}_{i=1}^N \left[\frac{|P_{i,g} - O_{i,g}|}{O_{i,g}} * 100 \right] \quad (g = 1, 2) \quad (10)$$

Samples that fell in and out of the polygon delineated by dual nitrate isotopes of sources were included in the performance assessment. Although samples that fell out of the polygon might reduce the performance of the model, it could check the applicability in different datasets. Practically, this is seldom the case that all samples fall in the polygon because it is very hard to find every source for the pollutant in the environmental study.

2. Results and discussion

2.1. Case 1 (three sources)

The study area of Case 1 is located in the Jinshoshan Karst World Heritage Site, Chongqing, China. Ming et al. (2020) determined the dual nitrate isotopes of groundwater ($n = 49$) and quantified the contribution of the three sources, including AP, MS, and SN, to groundwater nitrate. The coefficient of variation (CV) of $\delta^{15}\text{N}-\text{NO}_3^-$ and $\delta^{18}\text{O}-\text{NO}_3^-$ in Case 1 was 55.18 % and - 338.21 % (Table S2), respectively.

2.1.1. The contribution calculated by *MixSIAR* and *EMMTE*

As shown in Fig. 2 and Table S3, results of *MixSIAR* and *EMMTE* indicated that AP, MS, and SN contributed $2.67 \% \pm 2.11 \%$, $34.27 \% \pm 23.24 \%$, and $63.07 \% \pm 23.22 \%$, and $3.23 \% \pm 4.12 \%$, $32.72 \% \pm 32.44 \%$, and $64.05 \% \pm 31.84 \%$ to groundwater nitrate, respectively. The overall difference between the contribution calculated by the two models was insignificant ($p > 0.05$), indicating high overall consistency.

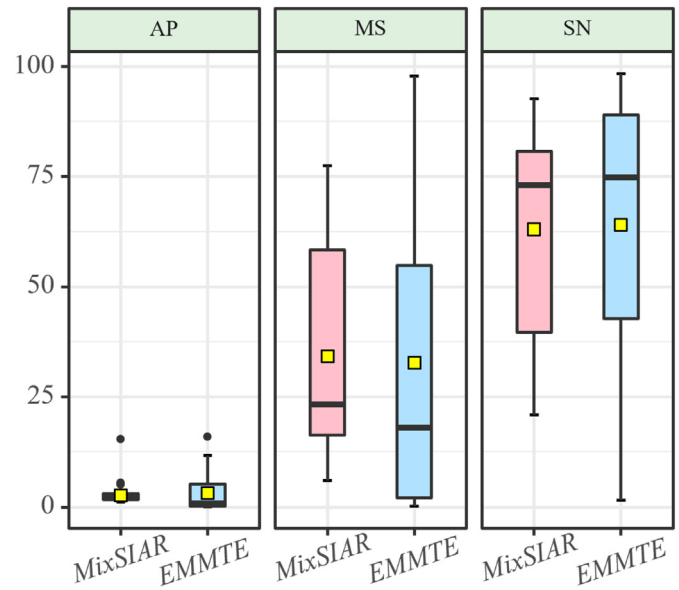


Fig. 2. Boxplot of the contribution of the various sources (AP – atmospheric precipitation, MS – manure and sewage, and SN – soil nitrogen) to nitrate in groundwater calculated by *MixSIAR* (pink) and *EMMTE* (blue) in Case 1. The upper point of the boxplot is the maximum and the lower is the minimum, the upper boundary of the box is the 75th percentile and the lower is the 25th percentile, the middle horizontal line is the median value and the middle square point is the average value. The significant differences of $p < 0.01$ and $p < 0.05$ between the two models are marked with ** and *, respectively. The insignificant differences are not marked with any symbols.

Moreover, there were good linear relationships between *MixSIAR* and *EMMTE* for the contribution of MS and SN as indicated by $R^2 > 0.6$ ($p < 0.01$), but the poor linear relationship between *MixSIAR* and *EMMTE* for the contribution of AP might be caused by some outliers (Fig. S1). We further found that the differences between contributions calculated by *MixSIAR* and *EMMTE* ranged from 1.89 % to 60.10 % with the maximum average of $15.60 \% \pm 13.96 \%$ for MS (Table S3) indicating large uncertainty. It may attribute to (1) some samples exceed even far away from the polygon delineated by various sources in particular the source of MS (Fig. 3) (Begum et al., 2022; Xue et al., 2012); (2) the discrimination of $\delta^{18}\text{O}-\text{NO}_3^-$ signature were not remarkable between MS and SN.

2.1.2. The performance of *MixSIAR* and *EMMTE*

The performance of *MixSIAR* and *EMMTE* in Case 1 was evaluated in Table 2 and Fig. 4. The higher G value in *EMMTE* ($\delta^{15}\text{N}-\text{NO}_3^-$, 0.534; $\delta^{18}\text{O}-\text{NO}_3^-$, 0.846) than in *MixSIAR* ($\delta^{15}\text{N}-\text{NO}_3^-$, 0.233; $\delta^{18}\text{O}-\text{NO}_3^-$, 0.662) indicated the stronger reliability of *EMMTE* than *MixSIAR*. In comparison, the RMSE of dual nitrate isotopes in *EMMTE* ($\delta^{15}\text{N}-\text{NO}_3^-$, 2.632; $\delta^{18}\text{O}-\text{NO}_3^-$, 2.174) was lower than in *MixSIAR* ($\delta^{15}\text{N}-\text{NO}_3^-$, 3.378; $\delta^{18}\text{O}-\text{NO}_3^-$, 3.221), indicating the better overall prediction effect of dual nitrate isotopes in *EMMTE*. Moreover, the RD range of dual nitrate isotopes in *EMMTE* was smaller than in *MixSIAR*, and the MRD in *EMMTE* ($\delta^{15}\text{N}-\text{NO}_3^-$, 7.17 %; $\delta^{18}\text{O}-\text{NO}_3^-$, 5.22 %) was remarkably lower than in *MixSIAR* ($\delta^{15}\text{N}-\text{NO}_3^-$, 18.60 %; $\delta^{18}\text{O}-\text{NO}_3^-$, 60.94 %). Therefore, the accuracy of *EMMTE* was higher than *MixSIAR*.

2.2. Case 2 (four sources)

The study area of Case 2 is the Chaobai River catchment, which lies in Beijing, China (116°30' E - 117° E, 40°10' N - 40°30' N). The vadose zone has rough lithology and strong permeability, which is beneficial for SW to recharge and it may elevate the nitrate load in groundwater (He et al., 2021; Li et al., 2019; Zheng et al., 2020). Cao et al. (2022) measured the

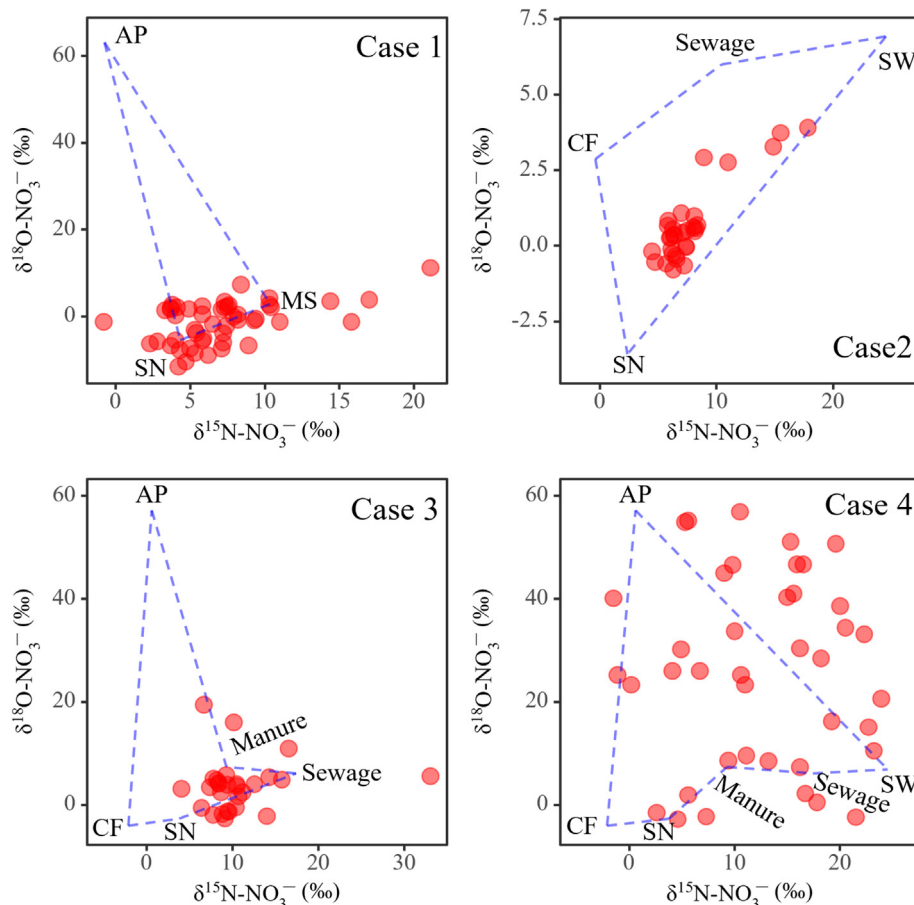


Fig. 3. The abundance distribution of dual nitrate isotopes for groundwater samples and various sources in four cases. AP – atmospheric precipitation, CF – chemical fertilizer, MS – manure and sewage, SN – soil nitrogen, and SW – surface water.

dual nitrate isotopes of groundwater ($n = 31$), and determined the contribution of four sources, including CF, sewage, SN, and SW, to nitrate. The CV of $\delta^{15}\text{N-NO}_3^-$ and $\delta^{18}\text{O-NO}_3^-$ in Case 2 was 39.14 % and 186.79 % (Table S2), respectively.

2.2.1. The contribution calculated by MixSIAR and EMMTE

As shown in Fig. 5 and Table S4, results of MixSIAR and EMMTE indicated that CF, sewage, SN, and SW contributed $10.91\% \pm 2.31\%$, $15.41\% \pm 3.37\%$, $53.77\% \pm 11.56\%$, and $19.91\% \pm 12.26\%$, and $10.87\% \pm 5.83\%$, $14.77\% \pm 6.19\%$, $53.86\% \pm 12.50\%$, and $20.50\% \pm$

14.35 % to groundwater nitrate, respectively. In addition, the contribution calculated by the two models was insignificant ($p > 0.05$, Fig. 5), indicating a high overall consistency. Moreover, there were good linear relationships between MixSIAR and EMMTE for the contribution of CF, SN, and SW as indicated by $R^2 > 0.6$ ($p < 0.001$) but a poor linear relationship for sewage ($R^2 = 0.23$, $p = 0.006$), which might be caused by some scatter points and outliers (Fig. S2). Furthermore, the differences in contributions calculated by the two models ranged from 0.06 % to 13.02 % as indicated by the maximum average of $4.46\% \pm 3.10\%$ for sewage (Table S4). The discrimination of $\delta^{18}\text{O-NO}_3^-$ signature between sewage and SW was relatively weak might be one reason (Fig. 3). However, the differences in contributions calculated by the two models were still lower than in Case 1. Hence, the accuracy and stability of EMMTE in Case 2 were better than in Case 1. It was explained that (1) all groundwater samples fell into the polygon delineated by source points (Fig. 3) (Derrien et al., 2018; Xue et al., 2012); (2) the discrimination of isotopic signature was relatively remarkable.

In a previous study by Ren et al. (2021) in the same study area, they found both SN and MS had a higher contribution and CF had a lower contribution, therefore, SN and MS were considered as dominant sources. However, the contribution of sewage was lower in Case 2, which was different from Ren et al. (2021). It may be attributed to the following two reasons: (1) groundwater has been recharged by a large amount of SW from many sources, including reclaimed water, reservoir water, and the clean water from the South-to-North Water Diversion Project and nitrate in sewage has been largely controlled in recent years (Castaldo et al., 2021; Li et al., 2019; Zhang and Yu, 2021; Zheng et al., 2020). (2) we only considered the source of sewage from Xian et al. (2016) but did not consider the source of manure. Thus, the source selection would definitely affect the source apportionment of nitrate.

Table 2

The performance of source apportionment by MixSIAR and EMMTE with dual nitrate isotopes in four cases.

Cases	Models	Tracers	G	RMSE	MRD (%)
Case 1	MixSIAR	$\delta^{15}\text{N-NO}_3^-$	0.233	3.378	18.60
		$\delta^{18}\text{O-NO}_3^-$	0.662	3.221	60.94
	EMMTE	$\delta^{15}\text{N-NO}_3^-$	0.534	2.632	7.17
		$\delta^{18}\text{O-NO}_3^-$	0.846	2.174	5.22
Case 2	MixSIAR	$\delta^{15}\text{N-NO}_3^-$	0.960	0.602	4.31
		$\delta^{18}\text{O-NO}_3^-$	0.965	0.236	14.64
	EMMTE	$\delta^{15}\text{N-NO}_3^-$	0.999	0.038	0.13
		$\delta^{18}\text{O-NO}_3^-$	0.999	0.004	0.09
Case 3	MixSIAR	$\delta^{15}\text{N-NO}_3^-$	0.511	1.941	13.58
		$\delta^{18}\text{O-NO}_3^-$	0.626	3.097	35.55
	EMMTE	$\delta^{15}\text{N-NO}_3^-$	0.306	2.313	0.24
		$\delta^{18}\text{O-NO}_3^-$	0.994	0.390	0.19
Case 4	MixSIAR	$\delta^{15}\text{N-NO}_3^-$	0.616	4.352	25.10
		$\delta^{18}\text{O-NO}_3^-$	0.386	14.260	38.69
	EMMTE	$\delta^{15}\text{N-NO}_3^-$	0.657	4.111	14.88
		$\delta^{18}\text{O-NO}_3^-$	0.786	8.416	17.36

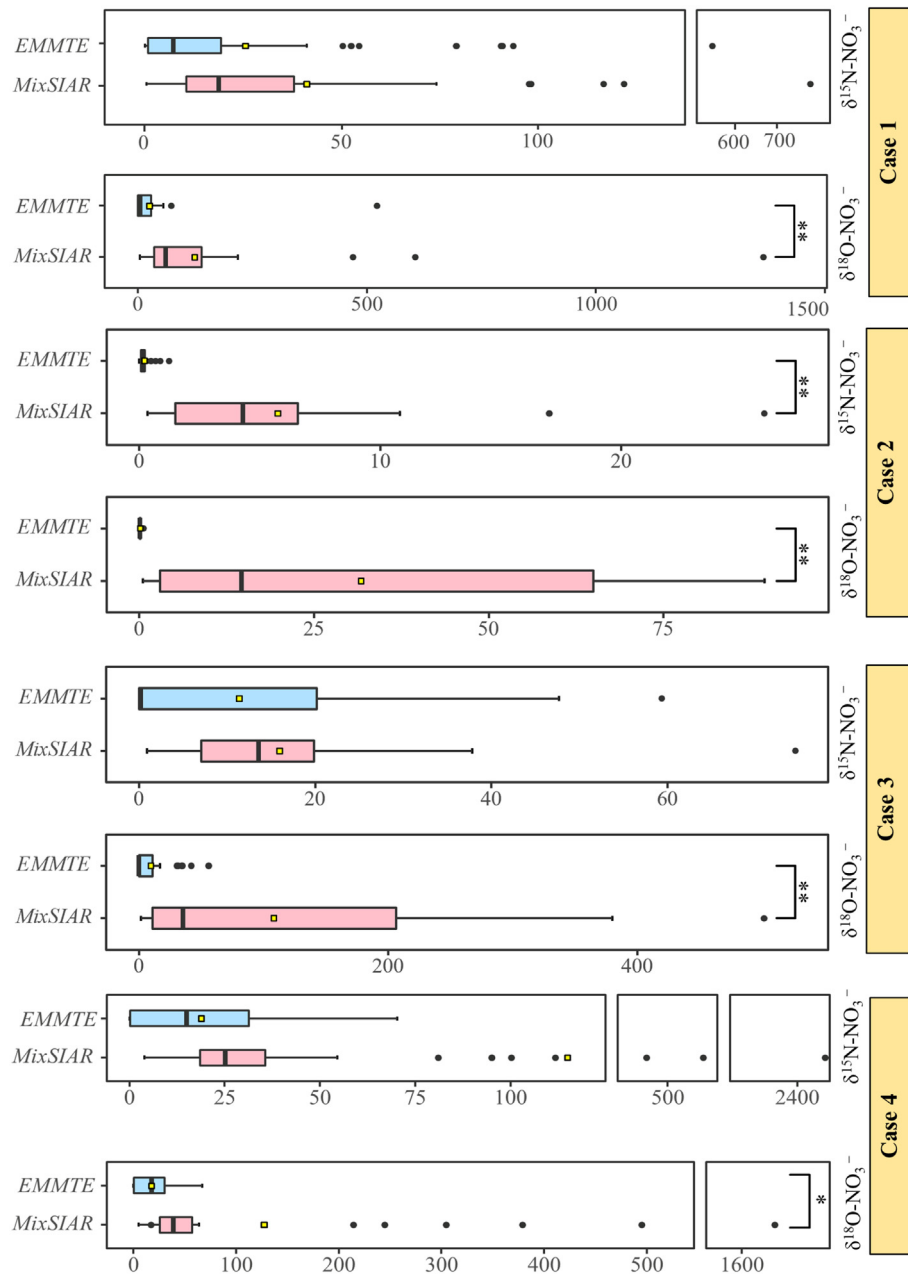


Fig. 4. The relative deviation (RD) (%) between O_{ig} and P_{ig} of dual nitrate isotopes in *MixSIAR* (pink) and *EMMTE* (blue). The significant differences of $p < 0.01$ and $p < 0.05$ between the two models are marked with ** and *, respectively. The insignificant differences are not marked with any symbols.

2.2.2. The performance of *MixSIAR* and *EMMTE*

As shown in Table 2 and Fig. 4, both *MixSIAR* and *EMMTE* had good reliability with the G value close to 1. The higher G value in *EMMTE* ($\delta^{15}\text{N-NO}_3^-$, 0.999; $\delta^{18}\text{O-NO}_3^-$, 0.999) than in *MixSIAR* ($\delta^{15}\text{N-NO}_3^-$, 0.960; $\delta^{18}\text{O-NO}_3^-$, 0.965) indicated that the reliability of *EMMTE* was stronger than *MixSIAR*. Moreover, the $RMSE$ of dual nitrate isotopes was small in the two models, indicating the overall prediction effect of both two models was good. In comparison, the lower $RMSE$ in *EMMTE* ($\delta^{15}\text{N-NO}_3^-$, 0.038; $\delta^{18}\text{O-NO}_3^-$, 0.004) than in *MixSIAR* ($\delta^{15}\text{N-NO}_3^-$, 0.602; $\delta^{18}\text{O-NO}_3^-$, 0.236) suggested the better overall prediction effect of *EMMTE*. The RD in *EMMTE* was significantly lower than in *MixSIAR* ($p < 0.01$) and its range in *EMMTE* was remarkably smaller than in *MixSIAR* (Fig. 4). Therefore, the performance of both *MixSIAR* and *EMMTE* was acceptable, and *EMMTE* was better than *MixSIAR* in Case 2.

2.3. Case 3 (five sources)

The study area of Case 3 is located in the northeast part of the Chengdu Plain, China ($104^\circ 15' \text{E} \sim 104^\circ 33' \text{E}$, $31^\circ 05' \text{N} \sim 31^\circ 20' \text{N}$). In the study by Zhang et al. (2020), the five sources of nitrate in groundwater ($n = 27$) including AP, CF, manure, sewage, and SN, were considered potential sources. The CV of $\delta^{15}\text{N-NO}_3^-$ and $\delta^{18}\text{O-NO}_3^-$ in Case 3 was 28.78 % and 143.89 % (Table S2), respectively.

2.3.1. The contribution calculated by *MixSIAR* and *EMMTE*

As shown in Fig. 6 and Table S5, results of *MixSIAR* and *EMMTE* showed that AP, CF, manure, sewage, and SN contributed $3.67 \% \pm 3.65 \%$, $10.37 \% \pm 3.35 \%$, $25.16 \% \pm 10.88 \%$, $30.34 \% \pm 14.80 \%$, and $30.45 \% \pm 11.25 \%$, and $3.60 \% \pm 6.84 \%$, $11.27 \% \pm 7.69 \%$, $12.27 \% \pm 10.20 \%$,

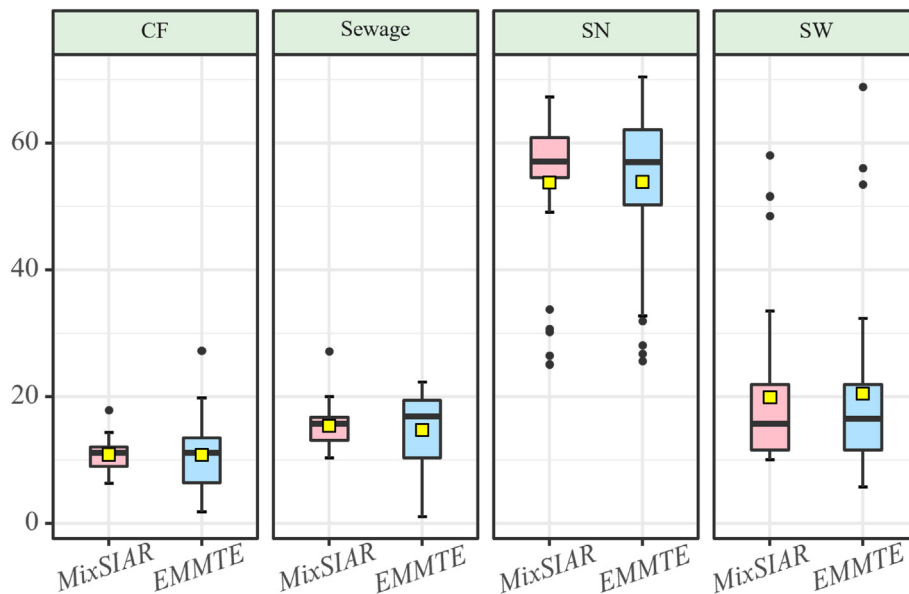


Fig. 5. Boxplot of the contribution of the various sources (CF – chemical fertilizer, sewage, SN – soil nitrogen, and SW – surface water) to nitrate in groundwater calculated by *MixSIAR* (pink) and *EMMTE* (blue) in Case 2. The upper point of the boxplot is the maximum and the lower is the minimum, the upper boundary of the box is the 75th percentile and the lower is the 25th percentile, the middle horizontal line is the median value and the middle square point is the average value. The significant differences of $p < 0.01$ and $p < 0.05$ between the two models are marked with ** and *, respectively. The insignificant differences are not marked with any symbols.

36.42 % $\pm$ 21.19 %, and 36.43 % $\pm$ 25.30 % to groundwater nitrate, respectively. The overall differences between the contribution determined by the two models were insignificant ($p > 0.05$) except for manure (Fig. 6), therefore the overall consistency was lower than in Case 1 and Case 2. Moreover, there were good linear relationships between *MixSIAR* and *EMMTE* for the contribution of AP, sewage, and SN as indicated by $R^2 > 0.55$ ($p < 0.001$), but poor linear relationships for the contribution of CF and manure (Fig. S3). It may be caused by the discrimination of $\delta^{18}\text{O}-\text{NO}_3^-$ in sources (CF and SN; manure and sewage) that were not obvious and some groundwater samples exceeded the polygon delineated by sources (Fig. 3) (Derrien et al., 2018; Xue et al., 2012). Furthermore, the

maximum differences between the contribution calculated by *MixSIAR* and *EMMTE* ranged from 0.31 % to 38.81 % with an average of 14.70 % $\pm$ 12.16 % for SN (Table S5) which was higher than in Case 2. Thus, the accuracy of *EMMTE* in calculating the contribution in Case 3 may be lower than in Case 2.

2.3.2. The performance of *MixSIAR* and *EMMTE*

In Case1 and Case 2, the G value of dual nitrate isotopes in *EMMTE* was higher than in *MixSIAR*. In Case 3, the G value of $\delta^{15}\text{N}-\text{NO}_3^-$ in *EMMTE* (0.306) was slightly lower than in *MixSIAR* (0.511), however, the G value of $\delta^{18}\text{O}-\text{NO}_3^-$ in *EMMTE* (0.994) was obviously higher than in *MixSIAR*

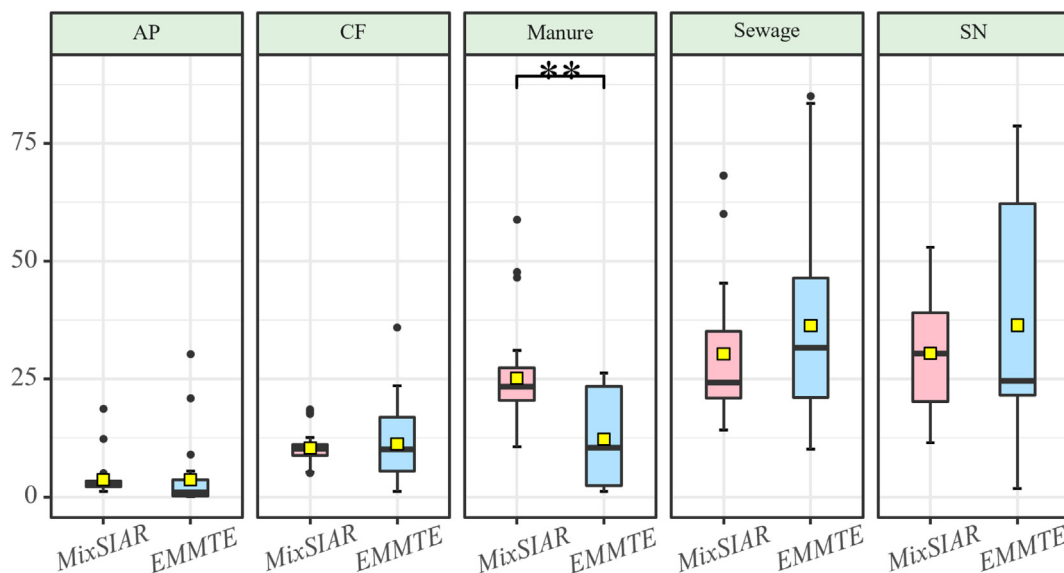


Fig. 6. Boxplot of the contribution of the various sources (AP – atmospheric precipitation, CF – chemical fertilizer, manure, sewage, and SN – soil nitrogen) to nitrate in groundwater calculated by *MixSIAR* (pink) and *EMMTE* (blue) in Case 3. The upper point of the boxplot is the maximum and the lower is the minimum, the upper boundary of the box is the 75th percentile and the lower is the 25th percentile, the middle horizontal line is the median value and the middle square point is the average value. The significant differences of $p < 0.01$ and $p < 0.05$ between the two models are marked with ** and *, respectively. The insignificant differences are not marked with any symbols.

(0.628) and closer to 1 (Table 2). Moreover, although the $RMSE$ of $\delta^{15}N-NO_3^-$ in $EMMTE$ (2.313) was slightly higher than in $MixSIAR$ (1.941), the $RMSE$ of $\delta^{18}O-NO_3^-$ in $EMMTE$ (0.390) was remarkably lower than in $MixSIAR$ (3.097) (Table 2). Furthermore, the MRD in $EMMTE$ ($\delta^{15}N-NO_3^-$, 0.24 %; $\delta^{18}O-NO_3^-$, 0.19 %) was clearly lower than in $MixSIAR$ ($\delta^{15}N-NO_3^-$, 13.58 %; $\delta^{18}O-NO_3^-$, 35.55 %) (Table 2 and Fig. 4). Overall, the performance of $EMMTE$ was better than $MixSIAR$ in Case 3.

2.4. Case 4 (six sources)

Through the first three practical cases, we concluded that $EMMTE$ was feasible in apportioning the contribution of various sources to groundwater nitrate, and its performance was better than $MixSIAR$. Moreover, to further evaluate the performance of the $EMMTE$, we generated 40 groups of virtual data on dual nitrate isotopes in Excel™ based on the range determined by the selected sources (AP, CF, manure, sewage, SN, and SW) involved in the first three cases (Table S1). The CV of $\delta^{15}N-NO_3^-$ and $\delta^{18}O-NO_3^-$ in Case 4 was 28.78 % and 143.89 % (Table S2), respectively.

2.4.1. The contribution calculated by $MixSIAR$ and $EMMTE$

As shown in Fig. 7 and Table S6, results of $MixSIAR$ and $EMMTE$ showed that AP, CF, manure, sewage, SN, and SW contributed $25.43\% \pm 16.32\%$, $9.04\% \pm 6.23\%$, $13.29\% \pm 7.46\%$, $17.28\% \pm 9.15\%$, $12.21\% \pm 10.86\%$, and $22.78\% \pm 14.63\%$, and $32.55\% \pm 23.11\%$, $11.84\% \pm 17.00\%$, $5.97\% \pm 6.49\%$, $7.28\% \pm 6.40\%$, $11.16\% \pm 16.18\%$, and $31.19\% \pm 21.34\%$ to groundwater nitrate, respectively. The overall difference between the contribution calculated by $MixSIAR$ and $EMMTE$ was insignificant for AP, CF, and SN ($p > 0.05$), but significant for manure, sewage, and SW ($p < 0.05$, Fig. 7), therefore, the overall consistency was lower than in other cases. Moreover, there were good linear relationships between the contribution calculated by $MixSIAR$ and $EMMTE$ for AP, CF, SN, and SW ($R^2 > 0.55$, $p < 0.001$), but poor linear relationships for manure and sewage (Fig. S4). The lower overall consistency and poor linear relationships might have a tight relationship with the extremely weak discrimination of $\delta^{18}O-NO_3^-$ signature among CF, manure, sewage, SN, and SW (Fig. 3). Furthermore, the differences in the contribution calculated by $MixSIAR$ and $EMMTE$ ranged from 0.34 % to 24.95 % as indicated by the maximum average of $12.76\% \pm 7.92\%$ for SW (Table S6), which was higher than in Case 2 but lower than in Case 1 and Case 3.

2.4.2. The performance of $MixSIAR$ and $EMMTE$

In Case 4, the G value of dual nitrate isotopes in $EMMTE$ ($\delta^{15}N-NO_3^-$, 0.657; $\delta^{18}O-NO_3^-$, 0.786) was higher than in $MixSIAR$ ($\delta^{15}N-NO_3^-$, 0.616; $\delta^{18}O-NO_3^-$, 0.386) and closer to 1 (Table 2), therefore the reliability of $EMMTE$ was stronger. Moreover, the $RMSE$ of dual nitrate isotopes in $EMMTE$ ($\delta^{15}N-NO_3^-$, 4.111; $\delta^{18}O-NO_3^-$, 8.416) was lower than in $MixSIAR$ ($\delta^{15}N-NO_3^-$, 4.352; $\delta^{18}O-NO_3^-$, 14.260) (Table 2), hence the overall prediction effect of $EMMTE$ was better. The MRD of dual nitrate isotopes in $EMMTE$ ($\delta^{15}N-NO_3^-$, 14.88 %; $\delta^{18}O-NO_3^-$, 17.36 %) was lower than in $MixSIAR$ ($\delta^{15}N-NO_3^-$, 25.10 %; $\delta^{18}O-NO_3^-$, 38.69 %) (Table 2 and Fig. 4), indicating $EMMTE$ had higher performance. Therefore, the overall performance of $EMMTE$ was better than $MixSIAR$.

2.5. The summary of $EMMTE$'s performance

Notably, groundwater samples in Case 2 all fell into the polygon delineated by sources and the discrimination of the isotopic signature among sources was relatively remarkable (Fig. 3). In the remaining three cases, there were some groundwater samples exceeded and even far away from the polygon delineated by its sources, therefore it implied that other potential nitrate sources might exist (Begum et al., 2022; Cao et al., 2022). In addition, the discrimination of isotopic signature in various sources was insignificant (the signature of $\delta^{18}O-NO_3^-$ between MS and SN in Case 1; the signature of $\delta^{18}O-NO_3^-$ among CF, manure, sewage, and SN in Case 3; the signature of $\delta^{18}O-NO_3^-$ among CF, manure, sewage, SN, and SW in Case 4) (Fig. 3). In summary, the better accuracy and performance depend on it that groundwater samples are all within the polygon delineated by sources and the discrimination of isotopic signature was remarkable among various sources (Derrien et al., 2018; Xue et al., 2012).

Among the four cases, the source apportionment results by $EMMTE$ had the highest consistency with $MixSIAR$ in Case 1 and 2 (Figs. 2 and 5), however, the consistency was very low for manure in Case 3 and manure, sewage, and SW in Case 4 (Figs. 6 and 7). It seemed that the number of sources might increase the inconsistency of source apportionments by $MixSIAR$ and $EMMTE$. Although the performance of $EMMTE$ was better than that of $MixSIAR$ as indicated by the larger G value and lower $RMSE$ and MRD (Table 2) and it was independent of the number of sources, the uncertainty of the contribution of various sources calculated by $EMMTE$ seemed to be larger than that of $MixSIAR$ as indicated by the wider box in the boxplot

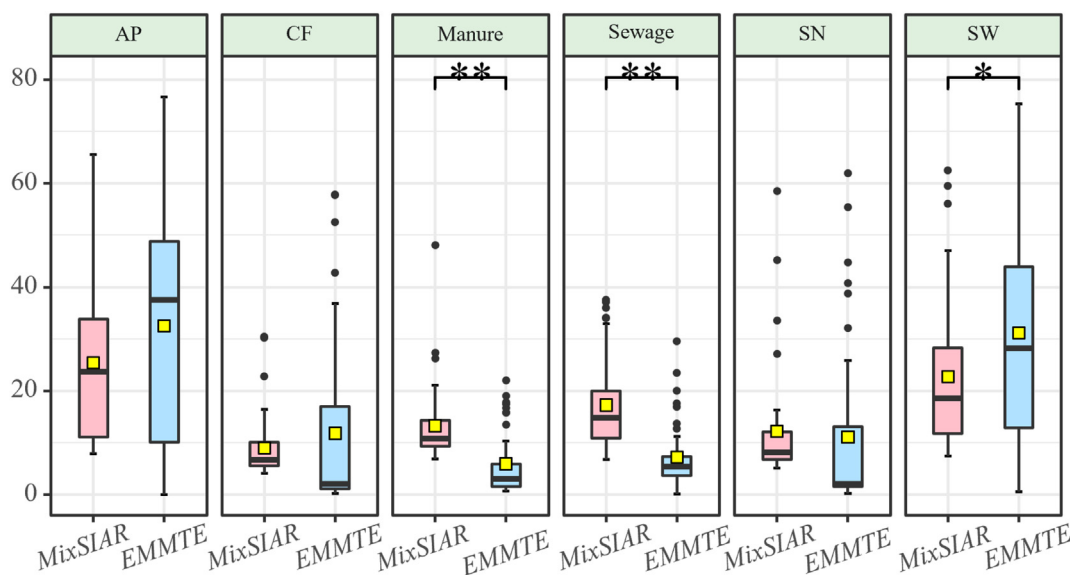


Fig. 7. Boxplot of the contribution of the various sources (AP – atmospheric precipitation, CF – chemical fertilizer, manure, sewage, SN – soil nitrogen, and SW – surface water) to nitrate in groundwater calculated by $MixSIAR$ (pink) and $EMMTE$ (blue) in Case 4. The upper point of the boxplot is the maximum and the lower is the minimum, the upper boundary of the box is the 75th percentile and the lower is the 25th percentile, the middle horizontal line is the median value and the middle square point is the average value. The significant differences of $p < 0.01$ and $p < 0.05$ between the two models are marked with ** and *, respectively. The insignificant differences are not marked with any symbols.

(Figs. 2, 5–7). Therefore, four random datasets ($n = 20$) of mixing fractions including three, four, five, and six sources according to the range of contributions in the four cases were generated (Tables S7–S10). Furthermore, the dual nitrate isotopes of samples were calculated by the randomly mixing fractions, and *EMMTE* and *MixSIAR* were used for source apportionment of the four randomly generated datasets. The absolute contribution differences in the four datasets by comparing the results from *EMMTE* and *MixSIAR* with the randomly generated fractions showed that the accuracy was independent of the number of sources (Tables S11–S14). It was worth noting that the accuracy of *EMMTE* was better than *MixSIAR* because absolute differences between the randomly generated and *EMMTE*-calculated contributions were significantly lower than ($p < 0.05$) absolute differences between the randomly generated and *MixSIAR*-calculated contributions for AP of the first dataset, sewage of the third dataset, and AP, manure, sewage, SN, and SW of the fourth dataset. There were no significant differences in accuracy between *EMMTE* and *MixSIAR* for the second dataset. Still, we also observed significantly larger absolute differences between the randomly generated and *EMMTE*-calculated contributions for MS of the first dataset, CF and SN of the third dataset, and CF of the fourth dataset.

As shown in Fig. S5, the predicted isotopic abundances by *MixSIAR* more deviated from the observed isotopic abundances than *EMMTE* in the four cases. In the comparison of the two isotopes, $\delta^{18}\text{O}-\text{NO}_3^-$ could discriminate the performance between *EMMTE* and *MixSIAR* as indicated by the significantly lower *RD* between the predicted and observed $\delta^{18}\text{O}-\text{NO}_3^-$ of samples by *EMMTE* ($p < 0.05$, Fig. 4). These phenomena could be observed in all four cases. However, $\delta^{15}\text{N}-\text{NO}_3^-$ could only discriminate the performance between *EMMTE* and *MixSIAR* in Case 2. Besides, $\delta^{18}\text{O}-\text{NO}_3^-$ in *MixSIAR* had higher *G* value, lower *RMSE* and *MRD* than in *EMMTE*, which further indicated *EMMTE* had better-predicting performance of $\delta^{18}\text{O}-\text{NO}_3^-$ than *MixSIAR* (Table 2). Furthermore, the *MRD* of $\delta^{18}\text{O}-\text{NO}_3^-$ in Case 1, Case 3, and Case 4 were remarkably higher than in Case 2, which may attribute to that the three former cases all took AP as a nitrate source with a high abundance of $\delta^{18}\text{O}-\text{NO}_3^-$ (Fig. 3). The higher constant of the multiple linear model (Eq. (7)) signified more contribution to the predicted isotopic abundance of $\delta^{18}\text{O}-\text{NO}_3^-$.

3. Conclusions

The advantages of *EMMTE* include a simple and user-friendly GUI, batch calculation, and visualization on the map compared to conventional models like *IsoSource*. After the comparative study with *MixSIAR* in various cases, the preliminary results showed that the contribution of various sources to groundwater nitrate calculated by *EMMTE* had high consistency with those by *MixSIAR*, indicating that *EMMTE* can be used for quantitatively evaluate the groundwater nitrate source apportionment. The performance of *EMMTE* seemed to be as good as *MixSIAR* as indicated by its higher *G* value, lower *RMSE*, and lower *MRD*. In the case that dual nitrate isotopes of groundwater samples were inside the polygon delineated by various sources (Case 2), the better performance of *EMMTE* was also highlighted. Moreover, *EMMTE* needs to be further optimized and improved in some aspects, including the variety of sources and the fractionation of tracers (isotopes).

In addition, *EMMTE* was developed based on the mass balance principle, therefore it could also apply to source apportionment of nitrate in other water bodies. If something (e.g., water, sulfate, organic matter, inorganic carbon, heavy metals, and trophic level) also obeyed the mass balance principle, *EMMTE* can also be used for source apportionment of it. For those objects, dual nitrate isotopes ($\delta^{15}\text{N}-\text{NO}_3^-$ and $\delta^{18}\text{O}-\text{NO}_3^-$) could be replaced with specific tracers. Furthermore, we also suggest more researchers use *EMMTE* to evaluate its performance and extend its application.

CRedit authorship contribution statement

Xu Cao: Data collection and collation, Software debugging, Visualization, Writing - Original draft preparation; **Wei He:** Writing - Reviewing & Editing, Funding acquisition, Software development; **Wei He:**

Conceptualization, Methodology, Resources, Supervision, Writing - Reviewing & Editing; **Yuanyuan Shi:** Conceptualization, Funding acquisition, Writing - Reviewing & Editing; **Tongyan An:** Writing - Reviewing & Editing; **Xiang Wang:** Visualization; **Fei Liu:** Methodology, Writing - Reviewing & Editing; **Yi Zhao:** Writing - Reviewing & Editing; **Pengpeng Zhou:** Writing - Reviewing & Editing; **Cuibai Chen:** Writing - Reviewing & Editing; **Jiangtao He:** Writing - Reviewing & Editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2023.161728>.

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